



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2026

PHYSICS (N530) **TIME: 45 MINUTES**

Candidates answer on the Question Paper.
Additional Materials: Mathematical set, Calculator

Specimen Paper 2026 — For examination practice only. Not an official MES paper.

READ THESE INSTRUCTIONS FIRST

- 1. Write your index number in the space provided above.
- 2. Write in dark blue or black ink. Do not use correction fluid.
- 3. You may use a soft pencil for any diagram, graph or rough working.
- 4. Diagrams are not drawn to scale unless otherwise specified.
- 5. Any rough working should be done in this booklet.
- 6. Answer **ALL** questions.
- 7. This document consists of **6** questions printed on **12** pages, numbered **2** to **13**.
- 8. The total marks for each question is shown in brackets () at the beginning of each question.
- 9. The number of marks for each part question is shown in brackets [].
- 10. The total number of marks for this paper is **50**.

For Examiners' use								
Question No.	1	2	3	4	5	6	Total	Signature
Examiner								
Team Leader								
CE/ACE								

Question 1 (10 marks)

Circle the correct answer. Each item carries one mark.

(a) Which of the following instruments is used to measure the **diameter** of a metal rod accurately?

- A A measuring tape
- B A ruler
- C A vernier caliper
- D A thermometer

(b) What is the S.I. unit of **time**?

- A hour (h)
- B minute (min)
- C second (s)
- D millisecond (ms)

(c) Fig. 1.1 shows four ray diagrams for reflection in a plane mirror. Which one correctly shows the **law of reflection**?

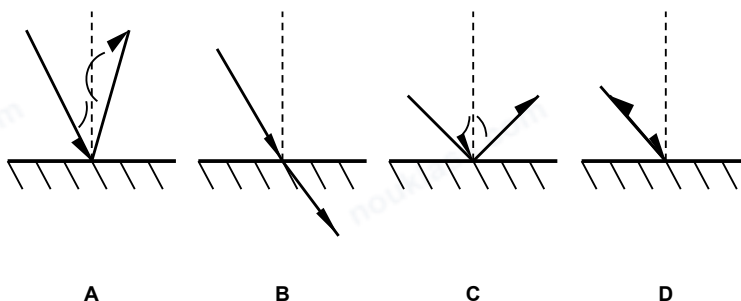


Fig. 1.1

(d) Which of the following is **NOT** a form of energy?

- A Kinetic energy
- B Force
- C Sound energy
- D Heat energy

(e) A lamp is rated at **60 W** and connected to a **240 V** supply. What is the **current** through the lamp?

- A 0.25 A
- B 4 A
- C 180 A
- D 14 400 A

(f) Which quantity is defined as the **rate of flow of charge**?

- A Potential difference
- B Resistance
- C Current
- D Power

(g) A ball is thrown upwards. At its **highest point**, which form of energy does it have?

- A Only kinetic energy
- B Only gravitational potential energy
- C Only heat energy
- D Both kinetic and gravitational potential energy

(h) Which of the following is a **non-luminous** object?

- A The Sun
- B A candle flame
- C A star
- D The Moon

(i) Which of the following correctly gives the relationship between resistance R , voltage V and current I ?

- A $R = VI$
- B $R = V / I$
- C $R = I / V$
- D $R = V + I$

(j) A stone falls from a height of 20 m. Which statement best describes the stone's energy **just before** it hits the ground? [Ignore air resistance]

- A It has only gravitational potential energy
- B It has only kinetic energy
- C It has equal amounts of GPE and kinetic energy
- D It has more GPE than kinetic energy

[10]

Question 2 (7 marks)

A girl stands **50 cm** in front of a plane mirror, as shown in Fig. 2.1. A vase is placed **120 cm** behind her, against the wall.

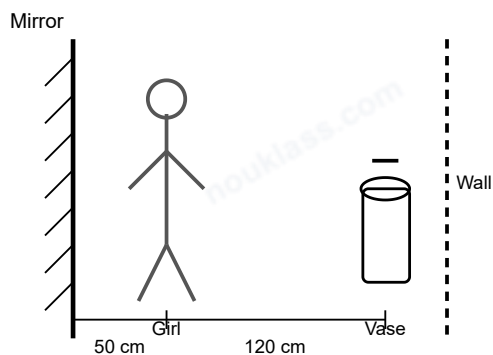


Fig. 2.1

(a) What is the distance of the **image of the girl** from the mirror?

Distance = cm [1]

(b) State **two** characteristics of the image of the girl formed in the plane mirror.

1.

2.

[2]

(c) The girl raises her **right hand**. Describe what the image appears to do.

[1]

(d) What is the distance between the girl and the image of the **vase** she sees in the mirror? Show all workings.

Distance = cm [2]

(e) The girl moves **10 cm** closer to the mirror. How far is her image from the mirror now?

[1]

Question 3 (5 marks)

Fig. 3.1 shows a ray of light passing from **air** into a glass block. The parts of the diagram are labelled **a** to **e**.

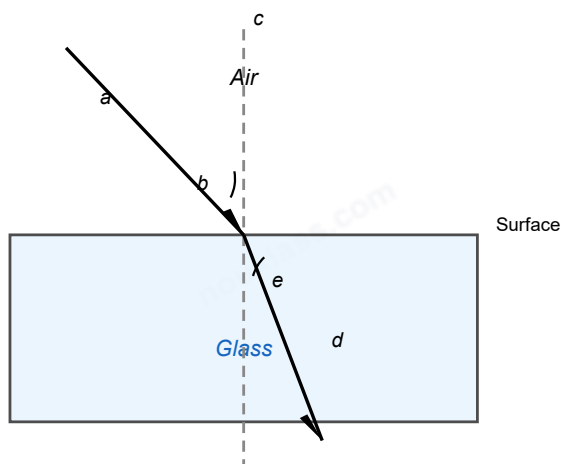


Fig. 3.1

Match each item in **Column A** with the correct label in Fig. 3.1.

Column A

- (a) Incident ray
- (b) Angle of incidence
- (c) Normal at the surface
- (d) Refracted ray
- (e) Angle of refraction

• •
• •
• •
• •
• •

Column B (labels in Fig. 3.1)

a
b
c
d
e

[5]

Question 4 (8 marks)

A student sets up an electric circuit to measure the resistance of two resistors **R1** and **R2** connected in series, as shown in Fig. 4.1.

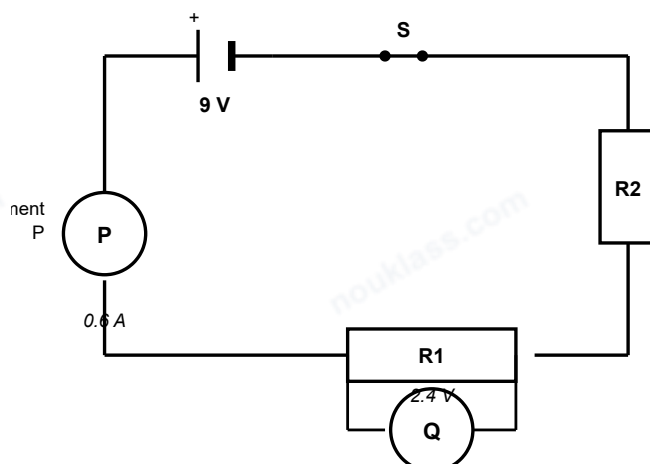


Fig. 4.1

(a) Instrument **P** reads **0.6 A** and Instrument **Q** reads **2.4 V**.
Name each instrument and state its function in the circuit.

Instrument P: Name Function: [1]

Instrument Q: Name Function: [1]

(b) Calculate the **resistance of R1**. Show all your workings.

Resistance of R1 = Ω [2]

(c) Calculate the **potential difference across R2**. Show all your workings.

Potential difference across R2 = V [2]

(d) Calculate the **resistance of R2**. Show all your workings.

Resistance of R2 = Ω [2]

6

Please turn over

Question 5 (10 marks)

(a) Table 5.1 lists some sources of energy.

(i) Put a tick (✓) beside **all** the **non-renewable** sources of energy in Table 5.1. [2]

Table 5.1

Energy source	Non-renewable (✓)
Diesel	
Wind	
Sun	
Coal	
Biogas	

(b) Match each description in **Column A** to the correct form of energy in **Column B**. [4]

Column A

Energy stored in food

Energy of a moving car

Energy stored in a compressed spring

Energy that enables us to see

• •

• •

• •

• •

Column B

Potential energy

Light energy

Heat energy

Kinetic energy

Chemical energy

(c) A coconut of mass **2 kg** falls from a tree that is **8 m** tall. [Take $g = 10 \text{ m/s}^2$]

(i) Calculate the **gravitational potential energy** of the coconut at the top of the tree. Show all workings.

Gravitational potential energy = J [2]

(ii) Assuming no energy is lost, state the **kinetic energy** of the coconut just before it hits the ground.

--

Kinetic energy = J [1]

(iii) Calculate the **speed** of the coconut just before it hits the ground. Show all workings. [KE = $\frac{1}{2}mv^2$]

Speed = m/s [1]

Question 6 (10 marks)

Fig. 6.1 shows the **distance-time graph** for a cyclist during a journey.

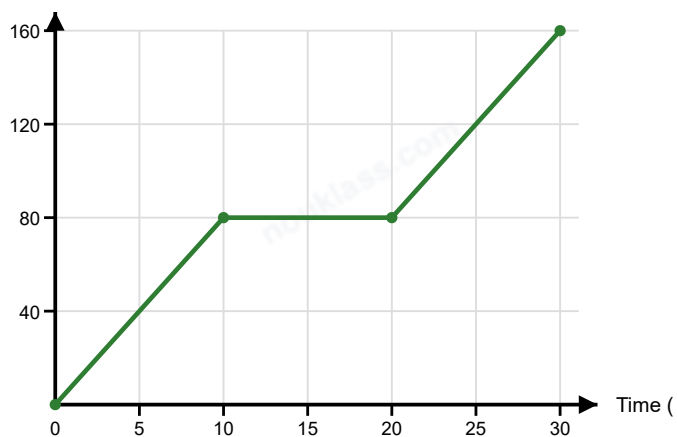


Fig. 6.1

(a)

(i) What is the **speed** of the cyclist between 0 and 10 s? Show all workings.

Speed = m/s [2]

(ii) Describe the motion of the cyclist between 10 s and 20 s.

[1]

(iii) What is the speed of the cyclist between 20 s and 30 s? Show all workings.

Speed = m/s [2]

(b) What is the **total distance** travelled by the cyclist in 30 s?

Total distance = m [1]

(c) Calculate the **average speed** of the cyclist for the complete 30 s journey. Show all workings.

Average speed = m/s [2]

(d) A different car is travelling at **25 m/s** and brakes uniformly to rest in **5 s**. Calculate the **deceleration** of the car. Show all workings.

Deceleration = m/s² [2]

Specimen Mark Scheme — Paper 2 of 2

Question 1 — Multiple Choice (10 marks, 1 each)

- (a) C — vernier caliper (measures diameter/small dimensions accurately)
- (b) C — second (s) is the S.I. unit of time
- (c) C — diagram showing equal angles of incidence and reflection on opposite sides of the normal, both above the mirror surface
- (d) B — Force is NOT a form of energy; it is a push or pull that can cause change in motion
- (e) A — 0.25 A $P = IV \rightarrow I = P/V = 60/240 = 0.25 \text{ A}$
- (f) C — Current (defined as Q/t , the rate of flow of charge)
- (g) B — Only gravitational potential energy (at the highest point, velocity = 0 so KE = 0)
- (h) D — The Moon (it does not produce its own light; it reflects sunlight)
- (i) B — $R = V/I$ (Ohm's law)
- (j) B — Only kinetic energy (just before hitting the ground, height ≈ 0 so GPE ≈ 0 , all energy is KE)

Question 2 — Plane Mirror (7 marks)

- (a) Image of girl is **50 cm** from the mirror (same distance as object). [1]
- (b) Any TWO of: virtual / same size as object / upright (erect) / laterally inverted / image is as far behind the mirror as the object is in front. [2]
- (c) The image appears to raise its **left hand** (lateral inversion). [1]
- (d) Image of vase in mirror:
Distance of vase from mirror = 50 (girl to mirror) + 120 (vase behind girl) = **170 cm**
Image of vase is 170 cm behind mirror.
Girl is 50 cm in front of mirror.
Distance from girl to image of vase = 50 + 170 = **220 cm**. [method [1], answer [1]]
- (e) Girl moves 10 cm closer $\rightarrow$ now 40 cm from mirror $\rightarrow$ image of girl is **40 cm** from mirror. [1]

Question 3 — Refraction Matching (5 marks)

- (a) Incident ray $\rightarrow a$ [1]
- (b) Angle of incidence $\rightarrow b$ [1]
- (c) Normal at the surface $\rightarrow c$ [1]
- (d) Refracted ray $\rightarrow d$ [1]
- (e) Angle of refraction $\rightarrow e$ [1]

Note: The refracted ray bends toward the normal when light travels from air (less dense) into glass (more dense medium).

Question 4 — Electricity (8 marks)

- (a) Instrument P: **Ammeter** — Function: measures the current flowing through the circuit [1]
Instrument Q: **Voltmeter** — Function: measures the potential difference (voltage) across R1 [1]
- (b) $R1 = V/I = 2.4/0.6 = \mathbf{4 \Omega}$ [formula [1], answer with unit [1]]
- (c) In a series circuit: $V_{\text{battery}} = V_{R1} + V_{R2}$
 $V_{R2} = 9 - 2.4 = \mathbf{6.6 \text{ V}}$ [method [1], answer with unit [1]]
- (d) $R2 = V_{R2}/I = 6.6/0.6 = \mathbf{11 \Omega}$ [formula [1], answer with unit [1]]

Verification: Total $R = 4 + 11 = 15 \Omega$, $I = 9/15 = 0.6 \text{ A}$ ✓

Question 5 — Energy (10 marks)

- (a)(i) Non-renewable: **Diesel** (✓) and **Coal** (✓) [1 mark each, total 2]
- (b) Matching [4 marks, 1 each]:
Energy stored in food $\rightarrow$ **Chemical energy**

Energy of a moving car → **Kinetic energy**

Energy stored in a compressed spring → **Potential energy**

Energy that enables us to see → **Light energy**

(c)(i) $GPE = mgh = 2 \times 10 \times 8 = 160 \text{ J}$ [formula [1], answer with unit [1]]

(c)(ii) $KE = 160 \text{ J}$ (by conservation of energy: GPE at top = KE at bottom, with no energy lost) [1]

(c)(iii) $KE = \frac{1}{2}mv^2 \rightarrow 160 = \frac{1}{2} \times 2 \times v^2 \rightarrow v^2 = 160 \rightarrow v = \sqrt{160} \approx 12.6 \text{ m/s}$

(Accept $4\sqrt{10} \text{ m/s}$ or 12.6 m/s) [substitution [0.5], answer with unit [0.5] — total [1]]

Question 6 — Distance-Time Graph (10 marks)

(a)(i) Between 0 and 10 s:

$$v = \Delta d / \Delta t = (80 - 0) / (10 - 0) = 80 / 10 = 8 \text{ m/s} \text{ [method [1], answer with unit [1]]}$$

(a)(ii) Between 10 s and 20 s: The cyclist is **stationary** (at rest). [1]

(a)(iii) Between 20 s and 30 s:

$$v = (160 - 80) / (30 - 20) = 80 / 10 = 8 \text{ m/s} \text{ [method [1], answer with unit [1]]}$$

(b) Total distance = **160 m** (read from graph at $t = 30 \text{ s}$) [1]

(c) $\text{Average speed} = \text{total distance} / \text{total time} = 160 / 30 = 5.3 \text{ m/s}$ (or $16/3 \text{ m/s}$)

Correct formula [1], answer with unit [1]

(d) $\text{Deceleration} = (v - u) / t = (0 - 25) / 5 = 5 \text{ m/s}^2$ (magnitude)

Formula [1], answer (positive value for deceleration) [1]

Accept: $a = -5 \text{ m/s}^2$ with explanation that the negative sign indicates deceleration.

** This specimen mark scheme is for practice purposes only. The real NCE 2026 paper and mark scheme are set exclusively by the Mauritius Examinations Syndicate (MES). This paper is produced by Noukclass.com and covers typical NCE Physics topics: measurement, light (reflection & refraction), motion (speed-time and distance-time graphs, acceleration), energy (GPE, KE, conservation, renewable vs non-renewable), and electricity (Ohm's law, series circuits, current, charge).*